

Exp No: 25

PRODUCER-CONSUMER PROBLEM USING SYSTEMC

PROGRAM

```
#include <systemc.h>

using namespace std;

// Producer Module
SC_MODULE(Producer) {
    sc_fifo_out<int> out;

    void produce() {
        for (int i = 1; i <= 5; i++) {
            cout << "Produced: " << i << endl;
            out.write(i);
            wait(1, SC_SEC);
        }
    }

    SC_CTOR(Producer) {
        SC_THREAD(produce);
    }
};

// Consumer Module
SC_MODULE(Consumer) {
    sc_fifo_in<int> in;

    void consume() {
        int value;

        while (true) {
            in.read(value);
            cout << "Consumed: " << value << endl;
            wait(1, SC_SEC);
        }
    }

    SC_CTOR(Consumer) {
        SC_THREAD(consume);
    }
};
```

```

// Main
int sc_main(int argc, char* argv[]) {
    sc_fifo<int> fifo(5);

    Producer producer("Producer");
    Consumer consumer("Consumer");

    producer.out(fifo);
    consumer.in(fifo);

    sc_start(10, SC_SEC);

    return 0;
}

```

OUTPUT

